

HOW TO PRODUCE YOUR PROJECT REPORT

Intended Learning Outcome

The objectives of CUH601CMD Software Engineering coursework are derived from the module objectives. Following the completion of the project, students are expected to be able to:

1. Evaluate an architectural design against accepted quality criteria.
2. Select, evaluate and use tools and techniques to successfully manage a large-scale software project, including configuration management and version control.
3. Use a range of appropriate tools to contribute to the development of a solution to a real-world problem.
4. Select, evaluate and apply standards, tools and techniques for assuring software quality.
5. Identify and engage with relevant social, professional, legal and ethical issues on a practical computing project.

The Project

The assessment for CUH601CMD Software Engineering is team-based. However, each team member will be assigned a relevant portion of individual tasks that will contribute to the completion of the team's project. Each team consists of 4 to 6 students, who will be working on the same project. The team must select one of the two study cases below.

Komodo Hub – A Digital Platform for Community-Supported Animal Conservation [Refer the module AULA page]

JustInTime – FloatFry's Manufacturing Resource Planning [Refer the module AULA page]

This module emphasizes a comprehensive software engineering approach to development. Therefore, the construction/coding aspect of the development of the proposed solution needs to be supported by equally strong evidence of practical project management and process control. To meet the objectives, students are asked to contribute to the creation of their team's comprehensive project report (~5000 words, citation excluded) covering three main aspects: **Process Control, Product Development, and Project Management**. Details on how to form and manage your team, how to manage your project including scheduling tasks and activities, the structure of the report document and what should be included in its content are available in the **HOW TO** documents provided on module AULA page.

Required Documentation

Either you are a Product Owner, Scrum Master, or Developer, you need to document your activities related to Process Control, Product Development, and Project Management. You must produce the documents listed in the table below every week, respectively to the role you are playing in the corresponding week. Every week, the Scrum Master in your team will collect all the documents produced by the other members. At the end of the project, the last Scrum Master will compile all ten weeks of documentation, complete it, and get it ready for submission. Your project report MUST follow the template available in the Report Structure section below.

EVENTS		ROLE		
		PRODUCT OWNER	SCRUM MASTER	DEVELOPERS
PROCESS CONTROL: all things related to how your team perform SCRUM appropriately, including Scrum meetings, Sprint Review, Sprint Retrospective, etc				
	SPRINT PLANNING	- Statement of product goal(s) – <i>start of Project</i> - Product Backlog – <i>start of Project and when there are changes</i>	Sprint Planning invitation, agenda, presentation, and report – <i>start of Sprint</i>	- Statement of Sprint goal(s) – <i>start of Sprint</i> - Sprint Backlog – <i>start of Sprint</i>
	WEEKLY SCRUM	Acknowledgement of weekly increments - <i>weekly</i>	Weekly Scrum invitation, agenda, presentation, and report - <i>weekly</i>	- Weekly increments report - <i>weekly</i> - Adjustment of Sprint Backlog – <i>weekly</i>
	SPRINT REVIEW	- Acknowledgement of test report – <i>end of Sprint</i> - Refinement of Product Backlog – <i>in between Sprints</i>	Sprint Review invitation, agenda, presentation, and report – <i>end of Sprint</i>	- Sprint increments report – <i>end of Sprint</i> - Test report– <i>end of Sprint</i>
	SPRINT RETROSPECTIVE	Concluding statement – <i>end of Sprint</i>	Sprint Retrospective invitation, agenda, presentation, and report – <i>end of Sprint</i>	

PRODUCT DEVELOPMENT: all things related to how your team design and develop the product, including analysing requirements, designing the software, modelling the design, etc				
	REQUIREMENT ANALYSIS	Requirement specifications – <i>start of Project and when there are changes</i>		
	DESIGN			- Business process analysis and models: Use Case Diagrams – <i>start of Project and when there are changes</i> - Business process analysis and models: Activity Diagrams – <i>start of Project and when there are changes</i> - Data analysis and models: DFDs – <i>start of Project and when there are changes</i> - Data analysis and models: ERDs/Class Diagrams – <i>start of Project and when there are changes</i> - Software architecture analysis and models: Class diagrams – <i>start of Project and when there are changes</i> - Hardware architecture analysis and models: Component diagrams – <i>start of Project and when there are changes</i> - System security analysis and design – <i>start of Project and when there are changes</i>

	DEVELOPMENT			- Interface prototype and codebase – <i>throughout development</i> - Database prototype – <i>throughout development</i>
	TESTING	Test verification points – <i>start of Sprint and when there are changes</i>		- Test cases, test data, test environment model, and test results – <i>weekly</i> - Test report – <i>end of Sprint</i>
PROJECT MANAGEMENT: all things related to how your team manage the project, including setting up milestones, scheduling tasks and activities, assigning jobs to members, etc				
	PROJECT DEFINITION	- Vision statement – <i>start of Project</i> - Mission statement – <i>start of Project</i> - Statement of scope and boundaries – <i>start of Project</i>	Risk Assessment – <i>start of Project</i>	
	SCHEDULING	Statement of milestones – <i>start of Project</i>	- Job allocation and management – <i>start of Sprint and when there are changes</i> - Project schedule – <i>start of Project and when there are changes</i>	
	MEETING ORGANIZATION		Meeting invitation, agenda, presentation, and report – <i>Scrum Meeting weekly, Sprint Review end of Sprint, Sprint Retrospective end of Sprint, others as needed</i>	
	MINUTES OF MEETINGS		- Minutes of meetings – <i>Scrum Meeting weekly, Sprint Review end of Sprint, Sprint Retrospective end of Sprint, others as needed</i> - Project final report – <i>end of Project</i>	

The Structure of Your Team's Project Report [on next page..]

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1.	Title Page							Sprint 2				
2.	Project Summary					11.		Sprint 2 Summary				
3.	Table of Content					12.		Week 1:	Product Owner Documentation			
4.	Project Schedule								Scrum Master Documentation			
5.	Job Allocation								Developer-1 Documentation			
	Sprint 1								Developer-2 Documentation			
6.	Sprint 1 Summary								Developer-3 Documentation			
7.	Week 1:	Product Owner Docu	mentation						Developer-4 Documentation			
		Scrum Master Docu	mentation			13.		Week 2:	Product Owner Documentation			
		Developer-1 Docume	ntation						Scrum Master Documentation			
		Developer-2 Docume	ntation						Developer-1 Documentation			
		Developer-3 Docume	ntation						Developer-2 Documentation			
		Developer-4 Docume	ntation						Developer-3 Documentation			
8.	Week 2:	Product Owner Docu	mentation						Developer-4 Documentation			
		Scrum Master Docu	mentation			14.		Week 3:	Product Owner Documentation			
		Developer-1 Docume	ntation						Scrum Master Documentation			
		Developer-2 Docume	ntation						Developer-1 Documentation			
		Developer-3 Docume	ntation						Developer-2 Documentation			
		Developer-4 Docume	ntation						Developer-3 Documentation			
9.	Week 3:	Product Owner Docu	mentation						Developer-4 Documentation			

Your team's project report should have a structure like the table on left.

IMPORTANT: On each and every document you produce individually, you **MUST** include information about your name, the role you are playing, and the date you are assigned to that role. Failure to do this will result in the marks not properly assigned to you.

			Scrum Master Documentation		15.		Week 4:	Product Owner Documentation
			Developer-1 Documentation					Scrum Master Documentation
			Developer-2 Documentation					Developer-1 Documentation
			Developer-3 Documentation					Developer-2 Documentation
			Developer-4 Documentation					Developer-3 Documentation
10.		Week 4:	Product Owner Documentation					Developer-4 Documentation
			Scrum Master Documentation		16.		Future Work	
			Developer-1 Documentation		17.		Conclusion	
			Developer-2 Documentation					
			Developer-3 Documentation					
			Developer-4 Documentation					